

B.Sc. in Computer Science and Engineering Thesis

Building a Semantic Search Engine using an Ontology: A Case Study on Social Media Influencers.

Submitted by

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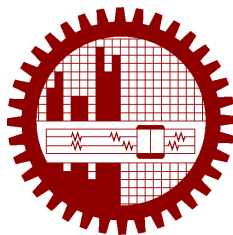
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CANDIDATES' DECLARATION

This is to certify that the work presented in this thesis, titled, “

Building a Semantic Search Engine using an Ontology: A Case Study on Social Media Influencers.”, is the outcome of the investigation and research carried out by us under the supervision of Dr. Muhammad Masroor Ali.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma or other qualifications.

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ABSTRACT

Social media influencers have become an essential part of many marketing initiatives due to their tremendous influence over their followers. Finding the right influencers for a given advertising campaign may be challenging and time-consuming because of the volume of content that is available on social media sites. Traditional keyword-based search methods frequently fall short of locating the most important influencers. To solve this issue, our ontology-based semantic search engine for social media influencers enables users to find influencers based on their audience's country, interaction rate, and category, more than just keyword matching. To create a semantic search engine that enables users to look for influencers, we built an ontology that defines concepts and relationships between social media platforms, their users, and the categorical content they produce. By incorporating our strategy into a social media analytics tool, we will be able to identify the appropriate influencers for a given campaign more precisely and effectively, demonstrating the efficacy of our approach. Our ontology-based approach for identifying influencer on social media sites is more intelligent and productive.

Chapter 1

Introduction

1.1 Background and Motivation

In the current marketing environment, social media influencers are becoming increasingly significant. The influence of people who have amassed sizable followings on social media platforms has grown along with the popularity of these sites. With corporations attempting to use influencers to market goods and services to their followers, they have become an essential component of many marketing efforts.

However, given the enormous amount of content available on social media platforms, finding appropriate influencers for a given campaign or product can be difficult and time-consuming. Traditional keyword-based search techniques frequently fall short of locating the most important influencers.

By building a semantic search engine using an ontology, we tried to create a more intelligent and effective approach to identify relevant influencers. An ontology-based method enables users to discover influencers based on a variety of characteristics beyond simple keyword matching, allowing for more accurate and nuanced search queries.

In this study, we developed an ontology that allows us to locate influencers based on the country of their audience, their interaction rate, and their category.

1.2 Research Objectives

The objective of the study is to create an ontology-based method for analyzing social media influencers. This ontology will provide as a framework for encoding the pertinent ideas, connections, and characteristics of social media influencers.

Using the created ontology, the study's ultimate goal is to construct a semantic search engine. From a variety of sources, including social media websites, news articles, and other

online materials, the search engine will be able to extract and process information about social media influencers. In order to give more precise and pertinent search results, the engine will employ the ontology to comprehend the meaning and context of the retrieved data. The overall goal of the ontology-based method and the semantic search engine is to offer a quicker and more accurate method of identifying and comprehending social media influencers. The method can save time and money by automating the process of data extraction and analysis while also offering insightful data and information to organizations, marketers, and researchers.

1.3 Research Questions

To achieve our goal, we will investigate the following research questions:

1. How can an ontology-based approach be used to model social platforms and influencers?
2. What are the key semantic relationships and attributes that should be included in the ontology to capture the relevant information about social media influencers?
3. What are the suitable technologies and tools for building a semantic search engine using the ontology and knowledge graph?
4. What are the benefits of using an ontology-based approach for social platform and influencer analysis compared to other methods?
5. How can an ontology for social platform and influencer analysis be integrated into existing social media analytics tools and platforms?
6. What are the limitations and challenges of using an ontology-based approach for social platform and influencer analysis, and how can they be addressed?

Our study will be guided by these research questions, which will also aid in achieving our study's goals. In the next chapter, we will learn about some terminologies, about the ontology and knowledge graph, and also about the search engine. In chapter three, we will demonstrate the construction of the ontology and semantic search and the integration of the ontology to a web tool. Chapter four will show the implementation details. We discussed about our study in chapter five. Finally we discussed about result and contribution in this field.

Chapter 2

Literature Review

2.1 Semantic Search Engines

2.1.1 Semantic Analysis

Even if search performance has greatly increased, not knowing the intents of the user when searching is undoubtedly holding us back. The process of gleaning meaning from text or voice data is known as semantic analysis, often referred to as semantic processing or semantic comprehension. In order to comprehend the intended meaning, it includes examining the structure, context, and relationships between words and sentences. The main objectives of semantic analysis are **Named Entity Recognition**, **Word Sense Disambiguation**, **Sentiment Analysis**, **Relation Extraction** etc. Techniques used in semantic analysis include **rule-based approaches**, **statistical models**, **machine learning**, and **deep learning**. Semantic dictionaries, ontologies, and large annotated datasets may be used in these methods for training and inference. We will use **rule-based approaches** on our **ontology** for our intended semantic analysis.

2.1.2 Semantic Search Engine and Key Features

Instead of relying merely on keyword matching, a semantic search engine attempts to comprehend the meaning behind search requests and the content of web pages. By taking into account the context, intent, and connections between words and concepts, it applies semantic analysis techniques to improve search results. Some Key features of semantic search engines:

- Entity Recognition and Extraction

- Conceptual Search
- Natural Language Understanding
- Contextual Understanding
- Sentiment Analysis
- Knowledge Graph Integration

Our target in this study is to incorporate structured knowledge graph of social media influencers with the search engine to enhance the understanding of relationships between entities and concepts.

2.1.3 Traditional Vs Semantic Search Engine

Traditional search engines mostly depend on keyword matching, where the results of the search are determined by the presence of particular terms in web pages. This method does not account for linguistic subtleties or the intended meaning behind a query, therefore it might not always produce correct answers. We'll use an example to illustrate the power of semantic search engines. Let a user searches the query "Top tiktok in facial care". Now keyword based search engine will look for the words "Top", "tiktok", "facial" and "care". But if we see how semantic search engine works, it will recognize the entity "tiktok" and "category" and then explore the relationships among them. Now category "facial care" should have annotations like "skin care", "body care" and it will provide individuals who has a tiktok account and hasInfluenceIn "facial care" or "skin care" or "body care". These results are not restricted by keyword only.

2.2 Ontology and Knowledge Graph

Ontology and Knowledge graph are closely related in the field of knowledge representation and semantic modelling. Let's see the core idea behind these.

A formal and explicit explanation of concepts and their connections within a specific domain or subject area is called an **ontology**. It offers a common vocabulary and comprehension of a domain and structures knowledge representation. Ontologies are commonly used in artificial intelligence, knowledge management, and the semantic web. They are typically represented using a formal language like **OWL** (Web Ontology Language) or **RDF** (Resource Description Framework).

The W3C Web Ontology Language (OWL) is a Semantic Web language created to convey detailed and intricate knowledge about various objects, collections of objects, and relationships between objects. The OWL ontology language is based on Description Logics, the family of class-based knowledge representation formalisms, allowing computer programs to use the knowledge expressed in OWL.

The Resource Description Framework (RDF) is a language for representing information about resources in the World Wide Web.

There are three main components to an ontology, which are usually described as follows:

- **Classes:** the distinct types of things that exist in our data.
- **Relationships:** properties that connect two classes.
- **Attributes:** properties that describe an individual class.

We can give an example:

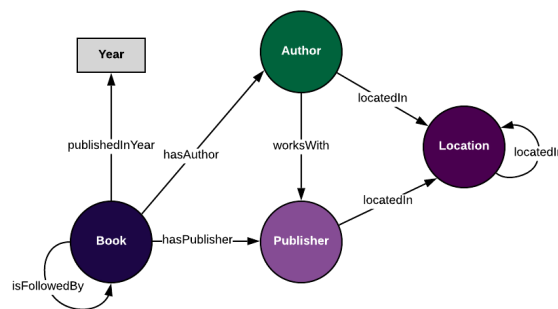


Figure 2.1: Ontology in a graph format, Image source: [1]

Here “Book”, “Publisher”, “Author”, and “Location” are classes of the ontology. “hasAuthor”, “hasPublisher”, “locatedIn” etc relations among the classes.

A knowledge graph is a particular kind of knowledge representation that organizes information as a network of nodes and edges, where nodes stand for entities (such people, places, or objects), and edges for the connections between those entities.

Therefore, a knowledge graph is a data structure that organizes and connects information about items and their interactions in a domain, whereas an ontology offers a structured vocabulary for describing entities and relationships in a particular domain.

However, a knowledge graph can be built using an ontology as a foundation. Information can be arranged and connected in a systematic fashion by using the concepts and relationships established in an ontology to construct nodes and edges in a knowledge graph. Similar to this, an ontology can be validated and improved using a knowledge graph by offering

instances and data from the actual world that can be used to evaluate and enhance the ontology's accuracy and completeness.

For the above ontology in the Figure 2.1, a knowledge graph could be:

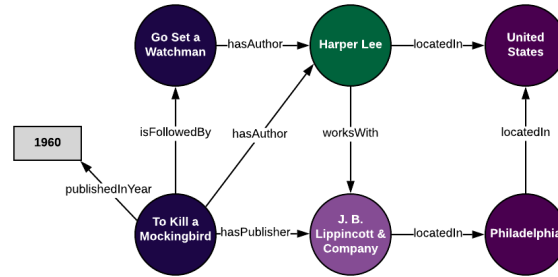


Figure 2.2: Knowledge Graph, Image source: [1]

Here “To Kill a MockingBird”, “Harper Lee”, “United States” etc are different type of entities of the ontology’s classes.

2.3 Reasoner for Ontology

In automated reasoning and knowledge representation, a reasoner is a piece of software that uses rules from an ontology to infer new knowledge from current knowledge. A reasoner works on a knowledge base defined in a formal language, such as OWL, and can conduct categorization, infer new relationships, verify for consistency, and respond to questions about the ontology. It has the ability to automatically infer logical conclusions from a set of axioms and decide whether a particular statement is implied by the axioms. Reasoners are employed in a wide range of applications, including the creation of ontologies, semantic web applications, knowledge-based systems, and intelligent agents.

Reasoning support is important for:

- Checking the consistency of the ontology and the knowledge
- Checking for unintended relationships between classes
- Automatically classifying instances in classes

2.4 Social Media Influencer Analysis

This study aims to comprehend how social media influencers affect the advertising business and investigate approaches to improve and speed up this process. Customers and their

choices are becoming more and more crucial in today's corporate world, and social media has emerged as a vital platform for advertising and brand promotion.

Social media influencers' eligibility, reach, engagement, and efficacy for brand partnership and marketing campaigns will be evaluated and assessed as part of the study. This study is crucial to ensuring that the selected influencers have an engaged and relevant audience, enabling brands to more successfully reach their target market.

The study has created an ontology to track engagement and performance metrics in order to achieve this. Metrics and market analysis help to reduce the dangers of working with influencers who lack actual audience engagement.

The study acknowledges that there is some risk associated with brands working with influencers because of how much of an impact influencers can have on a brand's reputation and image. Brands can choose an influencer for their advertising campaign with greater knowledge if they assess influencers utilizing analytics and analysis.

In general, this study aims to increase the efficiency of social media influencer marketing by offering a methodical evaluation and assessment of influencers. By doing this, it hopes to assist brands in having more success with their advertising and brand-promotion initiatives.

2.5 Existing Approaches and Technologies

Most of the paper worked on sentiment analysis of the contents posted by influencers. Analyzing content and videos is necessary for understanding influencing category of the influencers. But our focus were based on the semantic analysis for finding the desired influencers for brand collaboration or advertising. In the paper [2] shows how sentiment analysis can be used to find negative sentiment from contents. In this paper [3] social media data incorporated with ontology is used to analyze it and make a organizational decision. Using external data to decide on internal decisions. Our proposed approach can be used in this problem also but the specification is not fixed to a organizational decision. Our proposed ontology and implemented semantic search engine can provide much more flexibility and speed.

2.6 Tools and Libraries Used

The following tools were utilized in the development of the ontology and the semantic search engine:

2.6.1 Protégé

A free and open-source ontology editor and framework called **Protege** is used to create intelligent systems. In addition to using a variety of plug-ins, we created an OWL ontology using Protégé 5.5.0-win.

2.6.2 Pellet

An OWL 2 DL (Description Logic) reasoner is **Pellet**. It offers capability to compute the classification hierarchy, delineate inferences, validate ontology correctness, and respond to SPARQL requests.

2.6.3 Cellfie

A Protégé Desktop plugin for OWL ontologies that allows the input of spreadsheet data.

2.6.4 Neo4j

To create our knowledge graph, we used **Neo4j-1.5.6**. The knowledge graph was stored and queried using the graph database Neo4j. It was used to develop the semantic search engine and carry out knowledge graph searches.

2.6.5 Owlready2

Ontology-oriented programming is possible thanks to the Python module **Owlready2**. OWL 2.0 ontologies can be loaded as Python objects, edited, saved, and used with Hermit (included) for reasoning. Owlready2 allows transparent access to OWL ontologies (unlike the conventional Java-based API).

Chapter 3

Methodology

In this chapter we will discuss the methodology for our proposed ontology.

3.1 Domain and Scope of the Ontology

We have built this ontology from scratch. No ontology was reused as we couldn't find any suitable ontology for our case. Firstly we determined the domain and the scope of the ontology. The domain of the ontology for social platforms and influencers is social media, specifically social media platforms and the users who have significant influence on those platforms. The scope of the ontology includes defining the concepts and relationships between social media platforms, their users, and the content they produce. The ontology includes the following concepts:

- Social media platforms such as Twitch, Instagram, YouTube, and TikTok.
- Social media influencers.
- Including comments, likes, shares & viewers of a social platform
- Metrics used to measure social media performance, such as influence level.

Overall, the ontology for social platforms and influencers aims to provide a structured representation of the social marketing and its components.

We didn't generate the data by scrubbing. So we used a dataset from kaggle^[1].

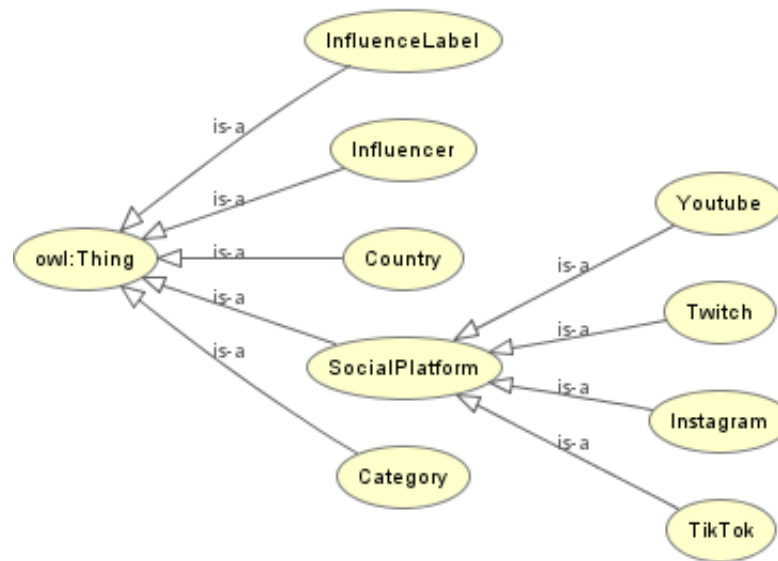


Figure 3.1: Class Hierarchy

3.2 Architecture and Hierarchy

We defined the classes and the class hierarchy. We set necessary terminologies. We used top down approach to make the class hierarchy. Top-down development process starts with the definition of the most general concepts in the domain and subsequent specialization of the concepts. The class hierarchy looks like below:

- owl:Thing
 - Category
 - Country
 - InfluenceLevel
 - Influencer
 - Social Platform
 - * Instagram
 - * TikTok
 - * Twitch
 - * Youtube

3.3 Class-specific properties

We started making the ontology using Protégé 5.5 which we will discuss in the tools section. We made the class hierarchy in a way so that if a new social platform come, we can easily

Table 3.1: Relationship Table

Class	Object Properties	Data Properties
Category	isInfluencedBy some Influencer	hasName only xsd:string
Country	None	hasName only xsd:string
InfluenceLevel	None	None
Influencer	hasAccount some SocialPlatform hasAudienceIn only Country	hasName only xsd:string
Social Platform	hasInfluencedLevel only InfluenceLevel	hasName only xsd:string hasFollower only xsd:integer
Instagram	Inherits all Social Platform's properties	hasAverageComments only xsd:integer hasAverageLikes only xsd:integer Inherits all Social Platform's properties
TikTok	Inherits all Social Platform's properties	hasAverageComments only xsd:integer hasAverageLikes only xsd:integer hasAverageShare only xsd:integer hasAverageViewers only xsd:integer Inherits all Social Platform's properties
Twitch	Inherits all Social Platform's properties	hasAverageViewers only xsd:integer Inherits all Social Platform's properties
Youtube	Inherits all Social Platform's properties	hasAverageComments only xsd:integer hasAverageLikes only xsd:integer hasAverageShare only xsd:integer hasAverageViewers only xsd:integer Inherits all Social Platform's properties

merge it into our ontology. We kept many common properties which will be inherited by the individual social platform. Then we defined all the necessary and sufficient properties of the classes. We used two types of properties. They are:

- **Object Properties:** These are relationships between two individuals.
- **Data Properties:** These are relations between an individual and a datatype such as xsd:string or xsd:dateTime.

Here are the properties of the classes in the ontology for social platforms and influencers:

In OWL the keyword **some** is used to denote existential restrictions. It means at least one.

The keyword **only** is used for universal restrictions. It means exactly one. This subsection responds to Research Question 1 and 2.

3.4 Facets of the Slots

We fixed the following aspects of the values the slot can accept: slot cardinality, slot-value type, domain, and range. These aspects describe the value type, permissible values, number of values, and other elements of the values the slot can take. After that, we at last had our ontology. There are 6 data properties, 7 object properties, and 9 classes. There are 24 SubClassOf axioms in total. Here are the facets of the slots we used in our ontology:

1. Category

- **isInfluencedBy:** A slot that links category to an influencer. With this we can find influencers influencing different categories.

2. Influencer

- **hasAccount:** A slot that links an influencer to a social platform account.
- **hasAudienceIn:** A slot that links an influencer to a country which has his/her most engaged viewers.

3. Social Platform

- **hasInfluenceLevel:** A slot that describes a Social account's influence level.

The data properties, such as `hasName`, `hasFollower`, etc, used in the ontology are intended to convey their lexical meaning and should be interpreted accordingly.

3.5 Semantic Search Engine

The goal of creating our ontology is to give semantic and robust support to the users who are looking for their brand collaborator or trying to increase their customer reach. For providing semantic support we started our journey to build an semantic search engine which will support our ontology. For creating a semantic search engine there are a numerous study and research. As the field is vast and the process is complex and iterative from different studies we tried to come up with a implementation that supports our use case. To focus on our case study we researched existing implementation of semantic search engine. Among the studies [4], [5], [6] our use case best suits [5]. In the study [5] they presented an approach for interpreting keyword queries using background knowledge available in ontologies. We mostly implemented their theory in our ontology to get semantic results from keywords. The implementation process and steps will be discussed in the implementation chapter. To understand the process, in this chapter we will look into some models in Ontology based Information retrieval and an illustrative example.

3.5.1 Ontology-Based Systems: Responding to User Queries

The process of ontology-based IR is shown in this section as an abstract framework. The user asks a question to the system, and the system responds with knowledge that has been formalized using a logical language. We concentrate on situations in which the query language used by the user does not correspond to the query language accepted by the system. For this reason, we outline the four models that make up our ontology-based Information Retrieval process as well as the underlying presumptions that guide our methodology in the appendix [A.1](#). Then, we outline a general method for converting a user inquiry into a formal system query.

3.5.2 Illustrative Example

To understand the four models let's move to our focus on a real life example. Where our user wants to retrieve all the influencers who are highly famous in the fashion category. This basically is information need, that a user has. We are identifying this as our Mental Model $\mathcal{O}_U$. We want to convert this information need to a system query that will retrieve these specific data.

Before diving into specific details we need to clarify on two assumptions while building this semantic search engine described in the [A.2](#)

This task is divided in three high level steps, which are

1. Elements in the user question $\mathcal{Q}_U$ are mapped to ontology elements from $\mathcal{O}_S$.
2. ontology elements are explored to better cover the initial information need in the mental model $\mathcal{O}_U$.
3. from this more refined ontological representation of the need, the query $\mathcal{Q}_S$ will be derived.

We are going to break down these three steps in higher detail by visualizing an illustrative example:

Step One — Mapping Terms to KB Entities

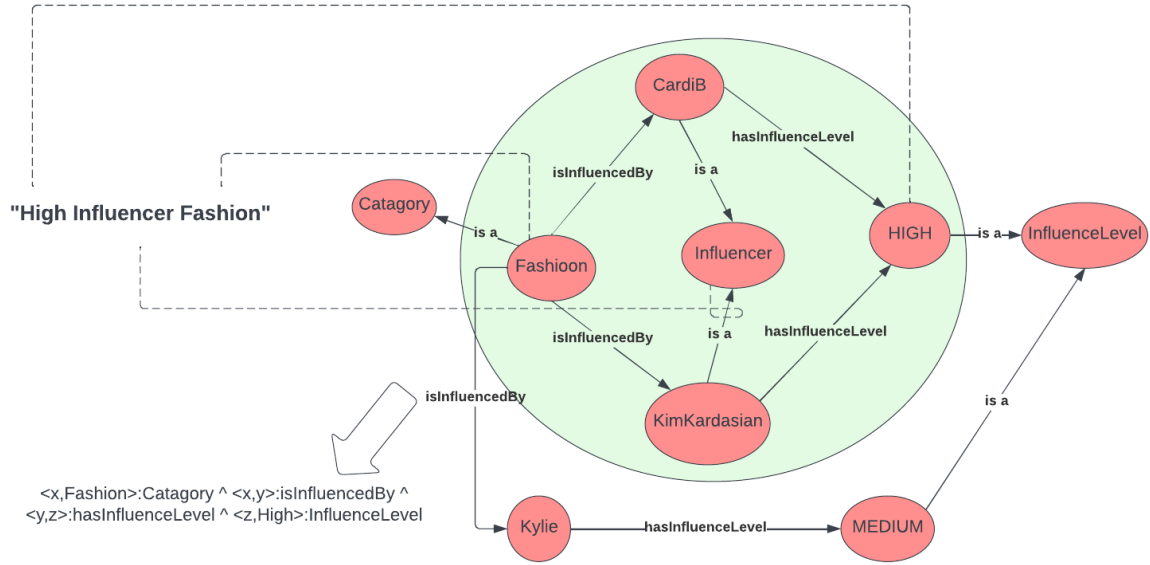
Due to our assumption A1, we are assuming that our user somewhat knows about the ontology and can search accordingly. If any user knows about the entity names and object properties or has a small understanding our Mental Model $\mathcal{O}_U$ can be converted to our User Query Model $\mathcal{Q}_U$. The refined query from the user could be “**High Influencer Fashion**”, which is

our $\mathcal{Q}_U$. In the first step we are going to map this keywords to our ontology entities if it exists. The mapping can be defined as a function f which maps elements of the user question model $\mathcal{Q}_U$ to entities of system resource model $\mathcal{O}'_S \subseteq \mathcal{O}_S$, i.e. $f: \mathcal{Q}_U \rightarrow \mathcal{O}'_S \subseteq \mathcal{O}_S$. For practical purposes, it is crucial that this function is **robust** in the sense that it also considers syntactic and spelling variants. Entities in the KB can be retrieved via their URIs. So the information need issued by the user is $\mathcal{Q}_U = \text{"High Influencer Fashion"}$. In our example we found three entities from our knowledge graph, High and Fashion is two individuals and Influencers is a concept in our Knowledge graph. These entities are the element of $\mathcal{O}'_S$, which is a subset of our System Resource Model $\mathcal{O}_S$. The elements of $\mathcal{O}_S$ are all the entities and relationships exist in the knowledge base. From above explanations three models described before are clarified. Remaining model System Query Model $\mathcal{Q}_S$ consists of the elements of the actual query which will be generated by the ontology based information retrieval process. However, these components still fall short of the user's information needs. Additionally, due to missing components, they still cannot be put together into a system query that produces the user's desired results. The items in the mental model that the user is aware of but does not specify correspond to these missing components such as **isInfluencedBy** between Fashion and influencer, **hasInfluenceLevel** between Fashion and High. These missing components are what we refer to as gaps, or parts of the mental model that the user does not explicitly state when expressing his or her information requirement.

Step Two — Exploring Connections Among KB Entities

After mapping the entities now we are going to explore these entities. Before starting the exploration procedure we will see some notation used in the research paper [5]. The notations used are described in the appendix A

Due to A1, we can restrict ourselves to the exploration of connections of the type $\langle i, C \rangle$, $\langle i_1, R, i_2 \rangle$, and $\langle i, U, j \rangle$. Due to A2, we are exploring upto a specific width d . We will follow the KB Exploration Algorithm of [5] for our exploration. As we have a concept Influencer we will take all the individuals of the concept and provide it to the I-P-I Traversal Algorithm of [5]. We have two more individual to explore which is Fashion and High. The exploration algorithm is basically a tuned iterative dfs in the knowledge graph. If already not visited we will visit the neighbouring entities by an edge which defines relationship among them, i.e. Object Properties. We will explore upto max depth of d and compute a subgraph. In this specific query we will get a connected subgraph. But there may be a case where the graphs are not connected. After the exploration we will have one or many subgraphs containing all the entities from $\mathcal{O}'_S$ upto depth d . But the problem here is that it will contain many information that doesn't corresponds to our user's information need. That's why we used Calculating Sub-Graphs of [5] to compute tree like graphs where the starting and finishing element should be

Figure 3.2: Workflow for the query "High Influencer Fashion", G_c

any entities present in the $\mathcal{O}'_S$. We are taking the edge following algorithm and constraining

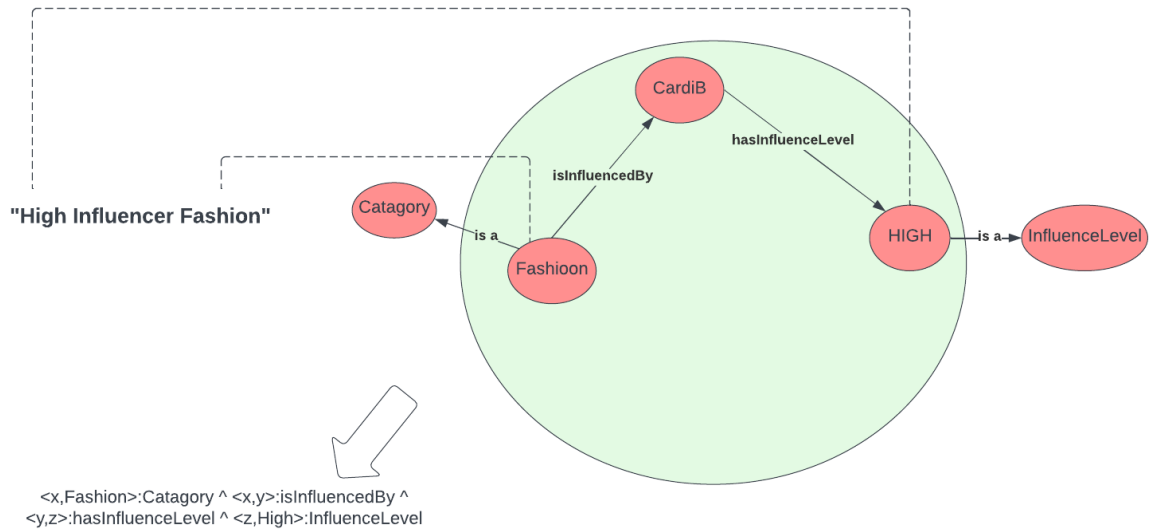


Figure 3.3: Workflow for the query "High Influencer Fashion"

the graph to contain Fashion at first and High at last containing influencer in the path. From this subgraph G_c we will now compute our query $\mathcal{Q}_S$

Step Three — Deriving DL Conjunctive Queries from Connections

For all subgraphs we need to derive a query and rank the resulting queries to get the most likely match of the initial question in the mental model of the user. To Map Connection to

queries:

- An edge in G_C , $type(v_i, v_c) \rightarrow \langle i, C \rangle \rightarrow \langle x, C \rangle$.
- An edge in G_C , $property_n(v_i, v_c) \rightarrow \langle i_1, R, i_2 \rangle and \langle i, U, j \rangle \rightarrow \langle x : y \rangle : R$.

The mapping from G_C to $\mathcal{Q}_S$ is complete because the investigation only includes these two categories of edges. In our example we have:

- $Category(uri1, Fashion) \rightarrow \langle x, Fashion \rangle : Catagory$
- $isInfluencedBy(uri1, CardiB) \rightarrow \langle x, y \rangle : isInfluencedBy$
- $hasInflunceLevel(CardiB, uri2) \rightarrow \langle y, z \rangle : hasInfluenceLevel$
- $influenceLevel(uri2, High) \rightarrow \langle z, High \rangle : InfluenceLevel$
- $\mathcal{Q}_S \rightarrow \langle x, Fashion \rangle : Catagory \wedge \langle x, y \rangle : isInfluencedBy \wedge \langle y, z \rangle : hasInfluenceLevel \wedge \langle z, High \rangle : InfluenceLevel$

Lastly we got our desired query set. From multiple G_C we could have multiple queries. That's why we need to rank these queries which would provide our efficient result for the information need. Ranking could be done by the connected distance of all the elements in $\mathcal{O}'_{\mathcal{S}}$. The more shorter the distance the more it satisfies the information need according to Assumption A2. Finally we got our query for the user.

Chapter 4

Implementation

4.1 Data Collection and Preprocessing

We have to preprocess the data to make it consistent with our ontology in order to use the “Social Media Influencers” dataset for our study. The top 1000 social media channels were collected in the dataset, which was got via Kaggle [7]. Each social media channel’s name, platform, URL, category, and audience demographics were included in the data, which was presented as an Excel file.

The preprocessing we have done are:

- Handled null values
- All the numerical values were given in M(million) and K(thousand) format. But we needed integer for our ontology. So we converted them into integers.

4.2 Instance Creation

In our study, we directly imported data from an xls file into our ontology using Protege’s Cellfie plugin. When compared to manually entering each instance, this method was quite effective and saved us a lot of time.

Cellfie automatically constructed instances for each row in the xls file when we mapped the columns of the spreadsheet to the traits of the relevant classes in the ontology. This made it unnecessary for us to manually enter each instance, allowing us to swiftly and effectively construct a large number of instances.

We used two types of query, DL(Description Logic) and SPARQL to query on the individuals. By creating instances and querying, we can demonstrate how the ontology can be used to

represent real-world data in the domain of social platforms and influencers.

4.3 Algorithm Designing for Influencing Level

An algorithm was designed to give the influencer appropriate influencing levels. We could have made the algorithm more accurate if we have real life data. We used some arbitrary weights on 'Likes', 'Comments', 'Shares' etc to get the level of the influencers. We could have utilized machine learning to determine which factors (followers, likes, comments, views, and shares) contribute the most to getting sponsorship if we had more information about the influencers' sponsorship rates. We used "owlready2" library to implement it in python.

- Inputs:
 - A list of social platforms and influencers.
 - The number of followers for each platform and influencer.
- Output:
 - A list of social platforms and influencers ranked by their influencing level.

4.4 Integration of the Ontology

There are numerous ways to include an ontology for social platform and influencer analysis into the platforms and tools used for social media analytics.

Strategy: 1 Creating APIs that let social media analytics software to access and use the ontology. This can be done by giving the analytics tool access to the ideas and relationships in the ontology using a web API. The analytics tool can then undertake a more thorough and accurate analysis of the social media data using these concepts and linkages.

Strategy: 2 Creating plug-ins or extensions that make use of the ontology is another option for integrating an ontology for social platform and influencer analysis into already-existing social media analytics tools and platforms. These add-ons or extensions can give the analytics tool new features like sophisticated filtering or visualization capabilities based on the concepts and relationships in the ontology.

We built a web tool with a simple UI. Then we made an API and followed the first strategy to integrate the ontology to the web tool.

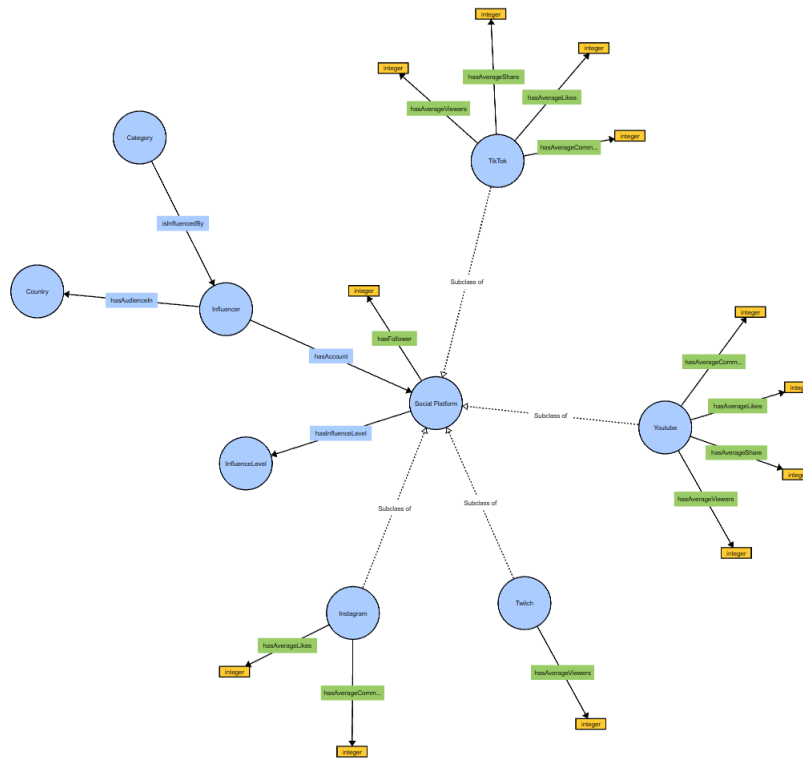


Figure 4.1: Ontology Diagram

4.5 Search Engine Implementation Details

As described in the methodology we tried to apply and improve the algorithms. We used “**owlready2**” library of python to explore and calculate subgraphs. While computing queries from subgraph our implementation provided some anomaly in the result. As building a semantic search engine is iterative and complex process, by giving more time and research we can refine our search engine. But we checked our compatibility of the ontology with a semantic search which could provide our objective.

Chapter 5

Discussion

5.1 Results and Analysis

The findings and analysis of our research on developing a semantic search engine employing an ontology for social media influencer analysis are presented in this part. We have finished building the ontology and successfully developed the ability to use Description Logic (DL) and a SPARQL query to find the right influencer for a marketing campaign.

The outcomes of our tests were encouraging, showing that the ontology can retrieve influencers based on particular standards and preferences. We were able to effectively filter and rank influencers based on many factors including audience demographics, engagement rates, specialty expertise, and social media presence by designing DL queries and utilizing the expressive capabilities of the ontology.

We were also able to run sophisticated graph-based queries on the ontology thanks to the integration of SPARQL query capabilities, providing more sophisticated and specialized influencer search operations. We might investigate the connections between influencers, find influencers with comparable audience characteristics or interests, and find unnoticed trends in the network of social media influencers.

This also showed good scalability and efficiency, handling significant amounts of social media data and delivering search results promptly.

5.2 Limitations and Challenges

There are several limitations and challenges associated with using an ontology-based approach for social platform and influencer analysis. Some of these include:

- **Complexity:** Since it calls for domain-specific knowledge and skills, creating and maintaining an ontology for social platform and influencer analysis can be a difficult and time-consuming effort.
- **Scalability:** Scaling an ontology-based method to accommodate the rising volume of data can be difficult when social platform data production keeps increasing.
- **Data Quality:** The efficiency of the analysis can be influenced by the precision and thoroughness of the data utilized to construct the ontology. Insights that are wrong or misleading can result from incomplete or inaccurate data.
- **Bias:** The accuracy and objectivity of the analysis may be impacted because the ontology may be influenced by the biases of its designers.
- **Integration:** It can be difficult to integrate the ontology with current social media analytics platforms and tools since it might necessitate major system changes.

It is critical to comprehend the ontology's constraints and potential biases, as well as the data sources and procedures used to create it, in order to resolve these issues. The ontology must be updated and maintained frequently to be current and correct. To increase the efficiency and scalability of the ontology-based method, it might also be essential to apply machine learning and other advanced analytics approaches.

5.3 Future Works

There are several possibilities for future work and potential upgrades to our system.

First, we can broaden the scope of our ontology to incorporate more elements that describe social media influencers, including the sentiment of their material or the demographics of their followers. This might make influencer searches more precise and efficient.

Second, by automating the classification and identification of influencers using powerful natural language processing techniques and machine learning algorithms, we can increase the precision and effectiveness of our approach.

Thirdly, we can combine our semantic search engine with a platform for managing social media to offer a more complete and efficient approach to influencer marketing. This might include tools like real-time performance tracking, automated campaign management, and influencer relationship management.

Finally, we can investigate the use of our semantic search engine in fields other than social media influencers, including e-commerce or online marketing. This could entail creating specialized search algorithms and user interfaces as well as modifying our ontology to capture the unique traits and connections of these domains.

Chapter 6

Conclusion

6.1 Conclusion

The Influencing Level is an important aspect of the ontology for Social Platforms and Influencers. In the section 4.3, we presented an algorithmic approach for designing the influencing level for the ontology. We also demonstrated the instance creation process for the influencing level using some example datasets. The ontology can be further extended by incorporating additional influencing factors, such as engagement rates and content quality, to provide a more comprehensive representation of social influencers.

Overall, the ontology for Social Platforms and Influencers can serve as a valuable tool for identifying influential individuals in various domains. The ontology can be used for targeted marketing. As the field of social media continues to evolve, the ontology can be updated and refined to reflect the changing dynamics of social platforms and influencers.

So now we have a fully working ontology where one user can find out the appropriate influencer for their marketing based on their budgets and categories and target audience.

6.2 Contributions to the Field

The goal of contemporary sentiment analysis research is to draw out the thoughts and emotions that are communicated in a text. Texts are divided into three categories of sentiment using text analysis, machine learning (ML), and natural language processing (NLP) techniques: “Positive,” “Negative,” or “Neutral.”

It’s crucial to remember that the ontology we’ve created is precisely intended to help you find the influencers who would work best with your advertising strategy. This ontology is distinct from the conventional sentiment analysis approach in that it focuses on social media influ-

encer analysis and categorization rather than text sentiment.

A complicated and iterative procedure is involved in building a semantic search engine that makes use of our ontology. Since it depends on the particular context and requirements, there is no one “correct” way to conduct semantic search. We used a methodology described in a research paper for our study that was in keeping with the ontology we created for our examination of social media influencers.

6.3 Implications for Industry and Academia

For social media influencer analysis, the creation of a semantic search engine based on an ontology has broad implications for both business and research.

- **Industry Implication:**

- **Enhanced Social Media Analysis:** For business experts engaged in social media analysis, the semantic search engine offers a potent tool. It makes it possible to more precisely and quickly identify pertinent information about social media influencers, which improves strategic planning and decision-making.
- **Improved Influencer Marketing:** By utilizing the ontology-based strategy, organizations may learn more about the traits, connections, and trends of influencers. This makes it easier to find influencers who are appropriate for marketing initiatives, leading to influencer marketing methods that are more effective and focused.
- **Competitive Advantage:** By utilizing the power of sophisticated data analysis methodologies, organizations who embrace and integrate the semantic search engine into their social media analytics tools and platforms get a competitive edge. They can find insightful information, comprehend consumer behavior, and come to data-driven conclusions.

- **Academic Implications:**

- **Advancement of Ontology Research:** By demonstrating an ontology’s usefulness in the context of social media influencer analysis, the study makes a contribution to the field of ontology creation and application. It increases our understanding of ontologies and how they might be used in real-world situations, especially in the field of social media analysis.
- **Bridging the Gap between Theory and Practice:** An effective illustration of how ontologies can be used to solve actual problems is provided by the case study on

social media influencers. It gives academics and researchers a real-world example to further investigate and hone ontology-based methods in various fields.

- **Collaboration Opportunities:** The research's conclusions provide opportunities for industrial and academic cooperation. Academic institutions can collaborate with businesses to improve the semantic search engine, test its effectiveness, and investigate other use cases in the area of social media analysis.

The implications mentioned above illustrate the significance and possible worth of developing a semantic search engine employing an ontology for social media influencer analysis. By utilizing the power of ontologies, business people may make better judgments, improve their marketing plans, and gain a competitive edge, while academics can deepen its understanding and further ontology research.

6.4 Final Remarks

In summary, the development of a Semantic Search Engine using an Ontology for social media influencer analysis represents a significant advancement in the field. This study has demonstrated the effectiveness of ontologies and semantic technologies in improving the accuracy and relevance of search results in social media platforms.

Despite our efforts, the performance of our semantic search engine in terms of accuracy and efficiency has not met our initial expectations. While we aimed to create a highly effective search tool that leverages the ontology for social platform and influencer analysis, we have encountered certain limitations and challenges in the development and implementation process.

Appendix A

Notations

- $\mathcal{Q}_U \rightarrow (k_1, k_2, \dots, k_n)$ where the k_i 's $\rightarrow$ keywords.
- $\mathcal{Q}_S \rightarrow$ grounded to DL conjunctive queries. conjunction of terms of the form $p : C$ or $\langle p, q \rangle : R$, where C is a concept, R is a role, and p, q are variables or individuals taken from $\mathcal{I}$ a set of individual names, or a set $\mathcal{V}$ of variable names.
- $\mathcal{O}_U \rightarrow$ corresponds to ontology entities in the disjoint union of the sets $\mathcal{T}$ (data values), $\mathcal{I}$ (individuals), $\mathcal{D}$ (data ranges), $\mathcal{C}$ (concepts), $\mathcal{R}$ (object properties), and $\mathcal{U}$ (data properties).
- Associations of $\mathcal{O}_U$ corresponds to associative interconnections of the types $\langle i, C \rangle$ $\langle i_1, R, i_2 \rangle$ and $\langle i, U, j \rangle$ where $i, i_1, i_2 \in \mathcal{I}, j \in \mathcal{T}, C \in \mathcal{C}, R \in \mathcal{R}$ and $U \in \mathcal{U}$

A.1 Semantic Search Engine Model

- **The Mental Model** $\mathcal{O}_U$
- **The User Question Model** $\mathcal{Q}_U$
- **The System Resource Model** $\mathcal{O}_S$
- **The System Query Model** $\mathcal{Q}_S$

A.2 Semantic Search Engine Assumption

1. **Assumption (A1) — Ontology-Mental Correspondence:**

- This assumption demands that the mental model $\mathcal{O}_U$ and the system resource model have a structural and entity-wise correspondence $\mathcal{O}_S$
- In other words, elements and the associative structure in $\mathcal{O}_U$ and $\mathcal{O}_S$, respectively, correspond to ontology entities and structure.

2. Assumption (A2) — Locality of Information Need:

- According to this assumption, the entities in the mental information need representation $\mathcal{O}_U$ that correspond to the ontology entities $\mathcal{O}'_S \subseteq \mathcal{O}_U$ must be connected over a distance d
- Suppose we have two ontology entities $a, b \in \mathcal{O}'_S$ there has to be a direct connection $\langle a, b \rangle$ or a sequence of x_i such that $a = x_0$ and $\langle x_0, x_1 \rangle, \langle x_1, x_2 \rangle, \dots, \langle x_{n-1}, x_n \rangle$, and $\langle x_n, b \rangle$ and $n \leq d$.

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